

IMPORTANT-NOTESCounting:

This develops some techniques for determining, without direct enumeration, the number of possible outcomes of a particular event or the number of elements in a set. Such sophisticated counting is sometimes called combinatorial analysis. It includes the study of permutations and combination.

BASIC COUNTING PRINCIPLE

There are two basic counting principles used throughout this chapter. The first one involves addition and the second one multiplication.

SUM RULE PRINCIPLE

Suppose some event E can occur in m ways, and a second event F can occur in n ways, and suppose both events cannot occur simultaneously. Then E and F can occur in $m + n$ ways.

PRODUCT RULE PRINCIPLE:

Suppose there is an Event E which can occur in m ways, and, independent of this event, there is a second event F which can occur in n ways. Then combinations of E and F can occur in mn ways.

The above principles can be extended to three or more events. That is, suppose an event E_1 can occur in n_1 ways, second event E_2 can occur in n_2 ways, and following E_i , a third event E_3 can occur in n_3 ways and so on. Then:

Sum Rule: If no two events can occur at the same time, then one of the events can occur in:

$$n_1 + n_2 + n_3 + \dots \text{ ways.}$$

PRODUCT RULE:

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If the events occur one after the other, then all the events can occur in the order indicated in,
 $n_1 \cdot n_2 \cdot n_3 \dots$ ways.

ex Suppose a college has 3 different history courses, 4 different literature courses and 2 different sociology courses.

- (a) The number m of ways a student can choose one of each kind of course is:

$$m = 3(4)(2) = 24.$$

- (b) The number n of ways a student can choose just one of the courses is:

$$n = 3 + 4 + 2 = 9.$$

There is a set theoretical interpretation of the above two principles. Specifically, suppose $n(A)$ denotes the number of elements in a set A . Then:

- 1) Sum Rule Principle = Suppose A & B are disjoint sets. Then

$$n(A \cup B) = n(A) + n(B).$$

- 2) Product Rule Principle =

Let $A \times B$ be the Cartesian Product of set A and B . Then

$$n(A \times B) = n(A) \cdot n(B).$$

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PIGEONHOLE PRINCIPLE

If 'n' pigeons are assigned to 'm' pigeonholes such that $(m < n)$ then atleast one pigeonhole must have 2 pigeons.

Extended pigeonhole principle

If 'n' pigeons are assigned to 'm' pigeonholes such that $(m < n)$ then atleast one pigeonhole contains

This means
 floor
 function
 Round up to
 smaller no.

$$\left\lceil \frac{n-1}{m} \right\rceil + 1 \text{ pigeons.}$$

Q1 If 13 students of a class are selected, Prove that atleast 2 of them ~~are~~ ~~will~~ will have birthday on the same month of a year.

Sol Here, 13 students ~~are~~ ($n=13$ pigeons) are selected from a class. There are 12 months ($m=12$ pigeonholes).
 $m < n$ pigeonhole principle applicable.

By pigeonhole principle atleast one pigeonhole contain more than 1 pigeons

$\therefore$ Atleast 2 students must have their birthday on the same ~~are~~ month of a year.

Q2 If 5 numbers are chosen from $\{1, 2, 3, \dots, 8\}$ then Prove that 2 of them will add upto 9.

Sol The number of the set $\{1, 2, \dots, 8\}$ can be grouped in pairs such that sum of each pair = 9 i.e.,
 $\{1, 8\}, \{2, 7\}, \{3, 6\}, \{4, 5\}$.

There are 4 groups $m=4$ (pigeon holes) and 5 numbers, $n=5$ (pigeons). **IMPORTANT-NOTES**

$$\Rightarrow m < n$$

pigeonhole principle is applicable.

By pigeonhole principle, one pigeonhole must have at least 2 pigeons.

$\therefore$ At least 2 numbers will add up to 9.

Q1 If 7 colours are used to paint 50 bicycles, then show that 8 of them must have same colour?

Sol Here $n = 50$ (pigeons)
 $m = 7$ (pigeonholes)

$$\therefore m < n$$

$\therefore$ Extended pigeonhole principle is applicable.

By Extended pigeonhole principle

$$\left\lceil \frac{n-1}{m} \right\rceil + 1 = \left\lceil \frac{50-1}{7} \right\rceil + 1 = \left\lceil \frac{49}{7} \right\rceil + 1 = 7 + 1 = 8$$

$\therefore$ Hence proved.

$\therefore$ that at least 8 bicycles have the same colour.

Q2 Show that if 30 dictionaries in a library contain a total of 61327 pages, then one of the dictionaries must have 2045 pages.

Sol Here $n = 61327$ pages (pigeons)
 $m = 30$ dictionaries (pigeonholes).

$$\therefore m < n$$

$\therefore$ Extended pigeonhole principle is applicable

By Extended Pigeonhole Principle **IMPORTANT-NOTES**

$$\left\lceil \frac{n-1}{m} \right\rceil + 1 = \left\lceil \frac{61327 - 1}{30} \right\rceil + 1 = \left\lceil \frac{61326}{30} \right\rceil + 1$$

2044.2

$$\left\lceil \frac{61326}{30} \right\rceil + 1 \Rightarrow 2044 + 1 = 2045$$

⑩ Hence proved

At least one dictionary will contain 2045 pages.

Q3 How many friends must have to guarantee that at least 5 of them will have birthday in the same month.

Sol Given $m = 12$ months (pigeon holes)
 $n = ?$

Also we have given the at least 5 of them will have birthday in the same month.

By Extended pigeonhole principle.

$$\left\lceil \frac{n-1}{m} \right\rceil + 1 = 5$$

$$\therefore \left\lceil \frac{n-1}{12} \right\rceil + 1 = 5$$

$$\Rightarrow n - 1 + 12 = 60$$

$$\Rightarrow n + 11 = 60$$

$$\Rightarrow n = 60 - 11$$

$$\Rightarrow n = 49$$

$\therefore$ There should be 49 friends.

Applications

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It is used in various fields to solve problems that involve distributing objects into containers. Here are some notable applications:

1. Computer Science: In computer science, the Pigeonhole Principle is used in hashing algorithms, where data items (keys) are assigned to hash values (buckets). The principle guarantees that if there are more items than buckets, collisions will occur, necessitating strategies to handle them.
2. Coding Theory: In coding theory, the Pigeonhole Principle helps in understanding error detection and correction. It implies that if you encode more messages than the number of unique codewords, some messages will share the same codeword, leading to potential errors.
3. Number theory: In number theory, it can prove the existence of certain properties within a set of numbers. For instance, it can show that within any set of $n+1$ integers, there exists a pair of integers whose difference is divisible by n .
4. Graph theory: In graph theory, ~~the~~ it is used to prove properties of graphs, such as the existence of a subgraph with certain characteristics.
For example, in any group of people, some subset of individuals will have mutual friendships or mutual non-friendships.

Permutations:

- The number of arrangements of n different things taken r ($\leq n$) at a time.
- The no. of ways in which r blank spaces can be filled up with n given things
- $${}^n P_r = \frac{n!}{(n-r)!}$$
- The order of things matters.

Ex How many numbers of 2 digit can be formed with the digits 1, 2, 3, if repetition of digits is not allowed?

Sol Here

$$\begin{aligned} \text{no. of digits } n &= 3 \\ r &= 2. \end{aligned}$$

$$\therefore \text{Using } {}^n P_r = \frac{n!}{(n-r)!}$$

$$\Rightarrow {}^3 P_2 = \frac{3!}{(3-2)!} = \frac{3 \times 2 \times 1}{1!} = 6$$

$\therefore$ 6 numbers of 2 digit can be formed.

Ex How many numbers lying between 100 and 1000 can be formed with the digits 1, 2, 3, 4, 5, if repetition of digits is not allowed?

Sol Here

$$\begin{aligned} \text{no. of digits } n &= 5 \\ r &= 3 \text{ (3 digits numbers)} \end{aligned}$$

$$\therefore {}^n P_r = \frac{n!}{(n-r)!}$$

$$= {}^5 P_3 = \frac{5!}{(5-3)!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{2!} = 60$$

$\therefore$ 60 numbers can be formed.

Q How many four digit numbers are there with distinct digits?

sol no. of digits $n = 10$ {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}
 $r = 4$ (4. digits numbers).

• ~~Q~~

$$P_n^r = \frac{n!}{(n-r)!} = \frac{10!}{(10-4)!} = \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{6 \times 5 \times 4 \times 3 \times 2 \times 1}$$

Now,

we have take make four digit numbers.

$$\underline{9} \quad \underline{9} \quad \underline{8} \quad \underline{7} : \rightarrow 9 \times 9 \times 8 \times 7 = 4536.$$

the first digits can be taken only 9 types.
 we cannot take 0 because then it will become 3 digit number.

$\therefore$ the first digit has 9.

Now,

the second digits also have 9 ways
 cause we can take 0 at second also.

the third have 8 and the fourth have 7

Q How many numbers are there b/w 100 & 1000 such that 7 is in the units place.

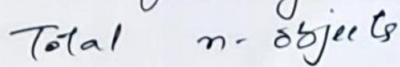
sol Between 100 & 1000 comes 3 digit numbers.
 and at units place we have 7 and also repetition is allowed.

$$\therefore \underline{9} \times \underline{10} \bullet \underline{7} = \text{~~900~~ } \text{~~900~~ }$$

~~900~~ members.

$\Rightarrow$ 90 choices.

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p objects are of one kind

9 " " " 1st kind.
8 " " " 2nd kind.

no. of way, in which n things can be arranged = $\frac{n!}{p! q! r!}$

8) Find the no. of 7 letter words that can be formed using letters of the word "BENZENE".

Sol $n = 7$. (~~no~~ Total number of letters).

Now .

$p \rightarrow N \rightarrow 2$ (In benzene).

$$2 \rightarrow 6 \rightarrow 3 \quad (10 \quad 11)$$
 $\mathcal{B} \rightarrow I$ $2 \rightarrow 1$

$0 \rightarrow 3$
 $1 \rightarrow 1$
 $2 \rightarrow 1$

These two does not repeat so they are used.

$$i. \quad \frac{n!}{p! q!} = \frac{7!}{2! 3!} = \frac{7 \times 6 \times 5 \times 4 \times \cancel{3!}}{2! \times \cancel{3!}} = \frac{7 \times 6 \times 5 \times 4}{2}$$

420

Q3 Find the number n of distinct permutations that can be formed from all the letters of each word. = 420.

Ⓐ THOSE Ⓑ UNSUAL Ⓒ SOCIOLOGICAL

Sol (iv) Here $n=5$ (5 letters).

All 0 letters are di Herent

$$\therefore 5! = 120$$



(5) $n = 7$ (7 letters).

U words is 3 times
and all other words are only 1 time.

$$\therefore \frac{n!}{3!} = \frac{7!}{3!} = \frac{7 \times 6 \times 5 \times 4 \times 3!}{3!} = 6 \times 7 = 420$$

(6) $n = 12$ (12 letters).

O word is 3 times

I word is 2 times

C word is 2 times

L word is 2 times

All other words are only 1 time.

SOCIOLOGICAL

$$\therefore \frac{12!}{3! 2! 2! 2!}$$

$$= \frac{12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3!}{3! \times 2! \times 2! \times 2!}$$

$$= \frac{12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4}{2 \times 2 \times 2} = 9979200$$

$$= 122 \times 90 \times 110$$

$$= 122 \times 9900 = 1,227,800$$

Permutation with repetition allowed **IMPORTANT NOTES**

- n - different things
- r - things taken at a time
- ~~Rep~~ Repetition allowed.
- $\text{No. of ways} = n^r$

Q How many numbers of 3 digits can be formed with the digits 2, 4, 6 and 8 when a digit may be repeated any number of times?

Sol Here

$$\text{No. of digits } (n) = 4.$$

$$\text{form 3 digits number } (r) = 3.$$

Repetition is allowed.

$$\therefore \text{No. of ways} = n^r = 4^3 = 4 \times 4 \times 4 = 16 \times 4 = 64$$

Q In how many ways 3 prizes can be distributed among 4 boys when a boy can get any number of prizes?

Sol

Here

$$\text{No. of boys } (n) = 4.$$

$$\text{No. of prizes } (r) = 3.$$

Repetition is allowed.

$$\therefore \text{No. of ways} = n^r = 4^3 = 64$$

Q Three cards are chosen one after the other from a 52-card deck. Find the no. of way this can be done

- with replacement
- without replacement.



Q. When repetition is allowed.

Here No. of cards $(n) = 52$.

cards chosen $(r) = 3$

Now, repetition is allowed

$$\therefore \text{No. of ways} = n^r = 52^3 = \underline{\underline{140608}}$$

Q. Where repetition is not allowed.

Total cards = 52.

Cards chosen = 3.

$$\underline{52} \quad \underline{51} \quad \underline{50}$$

$$\therefore 52 \times 51 \times 50 = \underline{\underline{132600}}$$

WORD BUILDING PROBLEMS.

Q. How many permutations of the letters
"A, B, C, D, E, F, G, H" containing the string ABC?

Ans. Here we treat ABC as one.

$\therefore$ Total number of letters = 6.

$$\therefore \text{No. of permutations} = 6! = 720$$

Q. Find the no. of permutations that can
be had from the letters of the word
"DAUGHTER".



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(i) taken all the letters together.

Sol when taking all the letters together
 $\therefore$ permutations = ~~8~~ 8!

(ii) beginning with D.

Sol Here the first space is fixed with D.

D _ _ _ _ _

Now we have 7 choices

$\therefore$ permutations = 7!

(iii) beginning with D and ending with R.

Sol Here the first space and the last are fixed.

D _ _ _ _ _ R

now we have 6 choices.

$\therefore$ permutations = 6!

(iv) Vowels being always together.

Sol Here we have vowels.

AUE $\rightarrow$ these will be treated as 1

Now ~~remaining~~ words

No. of remaining words are = 5

$\therefore$ Total no. of words here will be 6.

Also we can arrange AUE $\therefore$ permutation for AUE = 3!

$\therefore$ permutations = 6! \times 3!

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(v) not all vowels are together.

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Here

when all the letters are together,
the permutation $= 8!$

when all the vowels are together.

i.e. no. of permutations $= 6! \times 3!$

Now,

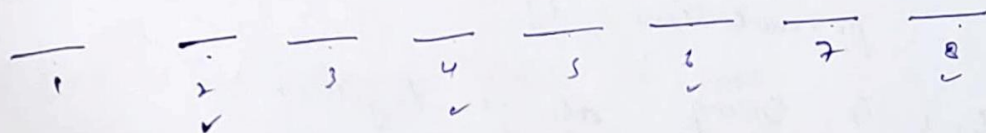
when all the vowels are not together =

Total permutations - permutation when
vowels are together.

$$\Rightarrow 8! - (6! \times 3!)$$

(vi) vowels occupying even places.

Imp



Total no. of consonants $= 5 = (D, L, H, T, R)$

i.e. permutations $= 5!$

even places $= 4 = n$

vowels $= 3 = r$

$$\text{i.e. Using } {}^n P_r = {}^4 P_3 = \frac{4!}{(4-3)!} = 4!$$

Total no. of permutations = $5! \times 4! = 120 \times 24$

= 2880

Q. How many different words can be formed with letters of the word ORIGINATE. so that (i) the vowels occupy odd places.

Sol. no. of consonants = 4 (R, D, N, T)
 i permutation for consonants = $4!$

Vowels = 4 (O, I, A, E).

Total no. of letters = 8.

No. of odd places = 4

i Here $n = 4$, $r = 4$

i Using ${}^n P_r = {}^4 P_4 = \frac{4!}{(4-4)!} = 4!$

i Total no. permutations = permutation of consonants $\times$ permutation of vowels.

$$\begin{array}{r} 5 \times 24 \\ 96 \\ 48 \times \\ \hline 576 \end{array}$$

$\Rightarrow 4! \times 4! = 24 \times 24 = 576.$

(ii) Beginning with O,

Here total no. of places = 8.

One will be occupied by 'O'.

i Remains places = 7

i permutation = $7!$



(c) Beginning with 0 and ending with 6.

Sol Here,

~~total~~ Total no. of places = 8.

Beginning and ending place that is 2 places are fixed.

∴ Remaining no of places = 6.

∴ No permutation = $6!$

COMBINATION.

→ Selection is made.

→ The ordering of the selected objects is immaterial (i.e., does not matter).

$${}^nC_r = \frac{n!}{r!(n-r)!} \quad \text{or} \quad C(n, r) = \frac{n!}{r!(n-r)!}$$

No. of combinations of n objects taken r at a time

$$\boxed{{}^nC_n = 1, \quad {}^nC_1 = n}$$

Q. $A = \{1, 2, 3\}$. Find the no. of subsets of set A having 2 elements.

Sol Here

no. of elements = $n = 3$

No. of elements taken at a time $(r) = 2$.

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Using $nCr = \frac{n!}{r!(n-r)!}$

$$\Rightarrow \text{no. of subsets} = \frac{3!}{2!(3-2)!} = \frac{3!}{2!1!} = \frac{3 \times 2!}{2!} = 3$$

Q. A farmer buys 3 cows, 2 goats and 4 hens from a man who has 6 cows, 5 goats and 8 hens. Find the number of choices that the farmer has.

Sol. Step of choices

Total number of cows $n = 6$.

No. of cows brought $(r) = 3$

$$\begin{aligned} \text{i. Combination for cows} &= nCr = \frac{n!}{r!(n-r)!} = \frac{6!}{3!(6-3)!} = \frac{6!}{3!3!} \\ &= \frac{6!}{3!3!} = \frac{6 \times 5 \times 4 \times 3!}{3! \times 3!} \\ &= \frac{6 \times 5 \times 4}{3 \times 3} = 20. \end{aligned}$$

ii. Total no. of goats $n = 5$
No. of goats brought $(r) = 2$

$$\begin{aligned} \text{i. Combination of goats} &= nCr = \frac{n!}{r!(n-r)!} = \frac{5!}{2!(5-2)!} = \frac{5!}{2!3!} \\ &= \frac{5 \times 4 \times 3!}{2! \times 3!} \\ &= \frac{5 \times 4}{2} = 10. \end{aligned}$$

iii. Total no. of hens $n = 8$
No. of hens brought $(r) = 4$

$$\begin{aligned} \text{i. Combination of hens} &= nCr = \frac{8!}{4!(8-4)!} = \frac{8!}{4!4!} \\ &= \frac{8 \times 7 \times 6 \times 5 \times 4!}{4! \times 4!} = 70. \end{aligned}$$



ii. Total number of choices = $20 \times 10 \times 10 = 1400$

Q A box contains 8 blue socks and 6 red socks. Find the number of ways two socks can be drawn from the box, if.

(a) they can be of any color?

Sol Here Total no. of socks (n) = $8 + 6 = 14$.

No. of socks drawn of any colour (r) = 2.

i. Total Combination = $14C_2$

$$\frac{14!}{2!(14-2)!} = \frac{14 \times 13}{2 \times 1} = 91$$

(b) they must be of the colour.

Sol Either there are 2 Blue or 2 Red.

Here Total no. of ~~red~~ blue socks (n) = 8.

No. of blue socks drawn (r) = 2.

i. Combination for blue socks = $8C_2 = 28$

ally Total no. of Red socks (n) = 6

No. of red socks drawn (r) = 2

i. Combination for Red socks = $6C_2 = \frac{6!}{2!(6-2)!}$

$$= \frac{6 \times 5 \times 4}{2 \times 1} = 15$$

i. Total number of combinations = $28 + 15 = 43$

8 Out of 5 men and 2 women a committee

of 3 is to be formed. In how many ways can this be done so as to include

(i) exactly one women.

Sol Total no. of women $n = 2$

No. of women taken $r = 1$

∴ Combination of women $= {}^2C_1$

Now, Total no. of men $m = 5$

No. of men taken when committee of 3

is to be made with one women $r = 2$

∴ Combination of men $= {}^5C_2$

Now,

we have taken one women and 2 men

∴ Total Combination $s = {}^2C_1 \times {}^5C_2$

$$= 2 \times \frac{5!}{2!(5-2)!} = \frac{2 \times 5 \times 4 \times 3!}{2 \times 3!}$$

(ii) Atleast one women.

$$= 20.$$

Sol Here, we can have one women out of 2 or both of two women.

So, the combination of women will be

2C_1 and 2C_2 .



when we will have one women
that time we will have 2 men
as the committee is made of 3.

is Combination of 2 men when one women
is present = 5C_2

→ when we will have 2 women that time
we will have to take 1 men.

is Combination of 1 men when 2 women are
present = 5C_1 .

Now,

Total no. of combination of men & women
for making a committee of 3.

$$= {}^2C_1 \times {}^5C_2 + {}^2C_2 \times {}^5C_1$$

$$= 20 + 1 \times 5$$

$$= 20 + 5 = 25$$

Principle of Mathematical Induction **IMPORTANT-NOTES**

→ It is a technique to prove mathematical statements involving positive integers.

Steps

- 1 | Let the given statement be denoted by $S(n)$.
- 2 | Take $n=1$ and verify that $S(1)$ is true. (i.e., take initial value).
- 3 | Assume that $S(k)$ is true.
- 4 | Verify that $S(k+1)$ is true by using step 3.
- 5 | $S(n)$ is true by principle of mathematical induction.

Ex 1 | Show that $1+2+3+\dots+n = \frac{n(n+1)}{2}$, by mathematical induction.

Sol | Let the given statement be denoted by $S(n)$.

Take $n=1$

$$S(1) = \frac{1(1+1)}{2} = \frac{1(2)}{2} = 1$$

$$\underline{1 = 1}$$

$S(1)$ is true.

Take $n=k$.

Assume that $S(k)$ is true,

$$\Rightarrow 1+2+3+\dots+k = \frac{k(k+1)}{2}$$

Now

$$n = k+1. \quad \text{~~ex 1~~}$$

$$\Rightarrow 1+2+3+\dots+k+k+1 = \frac{(k+1)(k+2)}{2}$$

we know that

$$1+2+3+\dots+k = \frac{k(k+1)}{2}$$

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~~$S(n)$ is true by principle of mathematical induction.~~

Show that $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n}{6} (n+1) (2n+1)$

$n \geq 1$ by M.I.

Sol. let the given statement be denoted by $S(n)$.

I. $n=1$.

$$1^2 = \frac{1}{6} (1+1) (2(1)+1)$$

$$1 = \frac{1}{6} (2) (3) \Rightarrow 1 = \frac{6}{6} \Rightarrow 1 = 1.$$

$\therefore S(1)$ is true.

II. Assume that $S(k)$ is true.

$$\Rightarrow 1^2 + 2^2 + 3^2 + \dots + k^2 = \frac{k}{6} (k+1) (2k+1). \text{--- (1)}$$

let $n = k+1$

$$\Rightarrow 1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2 = \text{--- (2)}$$

Using (1) we know that

$$1^2 + 2^2 + 3^2 + \dots + k^2 = \frac{k}{6} (k+1) (2k+1).$$

put (1) in (2)

$$\Rightarrow \frac{k}{6} (k+1) (2k+1) + (k+1)^2 = \frac{k+1}{6} (k+2) (2k+3)$$

$S(n)$ is true by principle of mathematical induction.

Q Show that $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n}{6} (n+1) (2n+1)$,
 $n \geq 1$ by MI.

Sol Let the given statement be denoted by $S(n)$.

I $n = 1$.

$$1^2 = \frac{1}{6} (1+1) (2(1)+1)$$

$$= 1 = \frac{1}{6} (2) (3) \Rightarrow 1 = \frac{6}{6} \Rightarrow 1 = 1.$$

is $S(1)$ is true.

II Assume that $S(k)$ is true.

$$\Rightarrow 1^2 + 2^2 + 3^2 + \dots + k^2 = \frac{k}{6} (k+1) (2k+1). \text{--- (1)}$$

III. Let $n = k+1$.

$$\Rightarrow 1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2 = \text{--- (2)}$$

Using (1) we know that

$$1^2 + 2^2 + 3^2 + \dots + k^2 = \frac{k}{6} (k+1) (2k+1).$$

put (1) in (2)

$$\Rightarrow \frac{k}{6} (k+1) (2k+1) + (k+1)^2 = \frac{k+1}{6} (k+2) (2k+3)$$

$$(k+1) \left[\frac{k}{6} (2k+1) + k+1 \right] = \frac{k+1}{6} (k+2) (2k+3)$$

$$\Rightarrow \frac{k+1}{6} [2k^2 + k + 6k + 6] = \text{'' ''}$$

$$\Rightarrow \frac{k+1}{6} [2k^2 + 7k + 6] = \text{'' ''}$$

$$\Rightarrow \frac{k+1}{6} [2k^2 + 4k + 3k + 6] = \text{'' ''}$$

$$\Rightarrow \frac{k+1}{6} [2k(k+2) + 3(k+2)] = \text{'' ''}$$

$$\Rightarrow \frac{k+1}{6} [(k+2)(2k+3)] = \text{'' ''}$$

$$\Rightarrow \frac{1}{6} (k+1)(k+2)(2k+3) = \text{'' ''}$$

$\therefore S(k+1)$ is true

$\Rightarrow$ By principle of mathematical induction

$S(n)$ is true.

Prove by Mathematical Induction. **IMPORTANT NOTES**

$$2+6+10+\dots+(4n-2) = 2n^2$$

Sol) Rept) Let the given statement be denoted by $S(n)$.

Step 2 Take $n=1$.

$$4(1)-2 = 2(1)^2 \Rightarrow 4-2 = 2 \Rightarrow 2=2$$

$\therefore S(1)$ is true.

Step 3 Assume that $S(k)$ is true.

$$\Rightarrow 2+6+10+\dots+(4k-2) = 2k^2 \quad \text{--- (1)}$$

Step 4 Let $n = k+1$.

$$\Rightarrow 2+6+10+\dots+(4k-2) + (4(k+1)-2) = 2(k+1)^2 \quad \text{--- (2)}$$

~~By~~ Comparing (1) and (2) we get.

$$2k^2 + (4(k+1)-2) = 2(k+1)^2$$

$$\Rightarrow 2k^2 + (4k+4-2) = 2(k+1)^2$$

$$\Rightarrow 2k^2 + 4k + 2 = 2(k+1)^2$$

$$\Rightarrow 2(k^2 + 2k + 1) = 2(k+1)^2$$

$$\Rightarrow 2(k^2 + 2 \cdot 1 \cdot k + 1^2) = 2(k+1)^2$$

$$\Rightarrow 2(k+1)^2 = 2(k+1)^2 \quad \left\{ \because (a+b)^2 = a^2 + 2ab + b^2 \right\}$$



$\therefore S(k+1)$ is true

$\therefore$ By Mathematical induction principle.

$$2 + 6 + 10 + \dots + (4n-2) = 2n^2.$$

Divisibility By Principle of MI.

Q. Prove by MI that $6^{n+2} + 7^{2n+1}$ is divisible by 43 for $n \geq 1$.

Sol. Let the given statement be $S(n)$.

Step 1 Take $n=1$

$$6^{1+2} + 7^{2(1)+1} = 6^3 + 7^3 = 216 + 343 = 559$$

$$= 43 \times 13.$$

$\therefore 43 \times 13$ is divisible by 43

$\therefore S(1)$ is true

Step 2 Let $S(k)$ be true.

$$6^{k+2} + 7^{2k+1} = 43m, \text{ for some integer } m.$$

Step 3

Take $n = k+1$

$$6^{(k+1)+2} + 7^{2(k+1)+1}$$

$$\Rightarrow 6^{k+3} + 7^{2k+3}$$

$$\Rightarrow 6 \cdot 6^{k+2} + 7^2 \cdot 7^{2k+1}$$

$$\Rightarrow 6(43m - 7^{2k+1}) + 49 = 7^{2k+1}$$

$$\Rightarrow 6 \times 43m - 6 \cdot 7^{2k+1} + 49 \cdot 7^{2k+1}$$

$$\Rightarrow 6 \times 43m + 43 \cdot 7^{2k+1}$$

$$= 43[6m + 7^{2k+1}], \text{ divisible by } 43.$$

$$\Rightarrow S(k+1) \text{ is true.}$$

$$\Rightarrow S(n) \text{ is true by principle of MI.}$$

Q Using MI, show that for all +ve integers n ,

$$7^{2n} + 16n - 1 \text{ is divisible by } 64.$$

sol let the given statement be denoted by $S(n)$.

$$\textcircled{1} \text{ Take } n=1$$

$$\Rightarrow 7^{2(1)} + 16(1) - 1 = 7^2 + 15 = 49 + 15 = 64$$

which is divisible by 64

$$\therefore S(1) \text{ is true.}$$

Assume $S(k)$ is true.

$$\rightarrow \cancel{7^{2k} + 16k - 1} \rightarrow 7^{2k} + 16k - 1 = 64m \text{ for some integer } m.$$

Now

$$n = k+1$$

$$\rightarrow 7^{2(k+1)} + 16(k+1) - 1$$

$$\rightarrow 7^{2k+2} + 16k + 16 - 1$$

$$\rightarrow 7^k \cdot 7^2 + 16k + 15$$

$$\rightarrow (64m - 16k + 1)7^2 + 16k + 15$$

$$\rightarrow 7^2 \cdot 64m - 16 \cdot 7^2 k + 49 + 16k + 15$$

$$\rightarrow 7^2 \cdot 64m - 16 \cdot 7^2 k + 16k + 64$$

$$\rightarrow 7^2 \cdot 64m - 16k(7^2 - 1) + 64$$

$$\rightarrow 7^2 \cdot 64m - 16k(48) + 64$$

$$\rightarrow 7^2 \cdot 64m - 12 \cdot 64k + 64$$

$$= 64(7^2 m - 12k + 1),$$

which is divisible by 64.

$\therefore S(n)$ is true, by P.O.I. 41

Using MI show that for all $n \in \mathbb{N}$

$11^n - 4^n$ is divisible by 7.

Sol. Let the given statement be denoted by $S(n)$

Take $n=1$.

$$\Rightarrow 11^1 - 4^1 = 11 - 4 = 7, \text{ which is divisible by } 7$$

Assume $n=k$ is true

$$\Rightarrow 11^k - 4^k = 7m, \text{ for some integer } m.$$

Take

$$n = k+1$$

$$11^{k+1} - 4^{k+1}$$

$$\Rightarrow 11^k \cdot 11 - 4^k \cdot 4$$

$\Rightarrow$ put the value of 11^k from (1)

$$(7m + 4^k)11 - 4^k \cdot 4$$

$$\Rightarrow 11 \cdot 7m + 4^k \cdot 11 - 4^k \cdot 4$$

$$\Rightarrow 11 \cdot 7m + 4^k (11 - 4)$$

$$\Rightarrow 11 \cdot 7m + 4^k (7)$$

$$\Rightarrow 7 \cdot 11m + 7 \cdot 4^k$$

$\Rightarrow 7(11m + 4^k)$,
which is divisible
by 7

$\therefore S(n)$ is ~~always~~ true
by P.O.M.I.



Prove Inequality by P of MI.

Q Show that $n^2 > 2n+1$ for $n > 3$ by MI.

Sol Let the given statement be denoted by $S(n)$

Take $n=3$.

$$3^2 > 2(3)+1$$

$$\Rightarrow 9 > 6+1 \Rightarrow 9 > 7$$

$\therefore S(3)$ is true.

Assume that $S(k)$ is true.

$$\Rightarrow k^2 > 2k+1$$

Now

Take $n = k+1$.

$$\Rightarrow (k+1)^2 > 2(k+1)+1$$

$$\Rightarrow k^2 + 2k + 1 > 2k + 2 + 1$$

$$\Rightarrow k^2 + 2k + 1 > 2k + 3$$

$$(k+1)^2 = k^2 + 2k + 1$$

$$> 2k+1 + 2k+1$$

$$\left\{ \because k^2 > 2k+1 \right\}$$

$$> 2k+1 + 2$$

$$> 2k+1 + 2$$

$$\left\{ \because 2k+1 > 2 \right\}$$

$$(k+1)^2 > 2k+3$$

$\therefore S(k+1)$ is true

$\therefore S(n)$ is true by P of MI.

Show that $(1 + \frac{1}{2})^n \geq 1 + \frac{n}{2}$ by M.I.

Sol Let the given statement be denoted by $S(n)$.

Take $n = 1$

$$\Rightarrow (1 + \frac{1}{2})^1 \geq 1 + \frac{1}{2}$$

$$\Rightarrow \frac{3}{2} \geq \frac{3}{2}$$

$$1 + \frac{k+1}{2}$$

$$\frac{k+3}{2}$$

$\therefore S(1)$ is true.

Assume that $S(k)$ is true.

$$\Rightarrow (1 + \frac{1}{2})^k \geq 1 + \frac{k}{2}$$

Now

Take $n = k+1$

$$\Rightarrow (1 + \frac{1}{2})^{k+1} = (1 + \frac{1}{2})^k (1 + \frac{1}{2})$$

$$\geq (1 + \frac{k}{2}) (1 + \frac{1}{2})$$

$$\geq (1 + \frac{k}{2}) (\frac{3}{2})$$

$$\geq \frac{3}{2} + \frac{3k}{2}$$

$$\geq \frac{3}{2} + \frac{k}{2}$$

$\therefore S(k+1)$ is true.

$\Rightarrow S(n)$ is true by P of M.I.

$\therefore \frac{3k}{2}$ is greater than $\frac{k}{2}$
So put $\frac{3k}{2}$ as $\frac{k}{2}$



BINOMIAL THEOREM.

IMPORTANT-NOTES

General form

$$(a+b)^n = {}^nC_0 a^{n-0} b^0 + {}^nC_1 a^{n-1} b^1 + {}^nC_2 a^{n-2} b^2 + {}^nC_3 a^{n-3} b^3 + \dots + {}^nC_n a^{n-n} b^n.$$

Q Expand $(3x+2y)^4$ by Binomial Theorem

$$\Rightarrow (3x+2y)^4 = {}^4C_0 (3x)^{4-0} (2y)^0 + {}^4C_1 (3x)^{4-1} (2y)^1 + {}^4C_2 (3x)^{4-2} (2y)^2 + {}^4C_3 (3x)^{4-3} (2y)^3 + {}^4C_4 (3x)^{4-4} (2y)^4$$

$$\Rightarrow {}^4C_0 (3x)^4 1 + {}^4C_1 (3x)^3 2y + {}^4C_2 (3x)^2 (2y)^2 + {}^4C_3 3x (2y)^3 + {}^4C_4 1 (2y)^4$$

$$= 1 (3x)^4 + \frac{4!}{1!(4-1)!} (3x)^3 2y + \frac{4!}{2!(4-2)!} (3x)^2 (2y)^2 + \frac{4!}{3!(4-3)!} 3x (2y)^3 + \frac{4!}{4!(4-4)!} (2y)^4$$

$$= (3x)^4 + 4(3x)^3 2y + 6(3x)^2 (2y)^2 + 4(3x)(2y)^3 + (2y)^4$$

$$= 81x^4 + 96x^3y + 72x^2y^2 + 48xy^3 + 16y^4$$

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

$${}^nC_0 = 1$$

$${}^nC_n = 1$$

$$\frac{n!}{r!(n-r)!}$$

$$= 81x^4 + 8 \times 27 \cdot x^3y + \text{IMPORTANT-NOTES} \cdot 16y^4$$

$$\Rightarrow 81x^4 + 216x^3y + 216x^2y^2 + 96xy^3 + 16y^4$$

Q Expand $(99)^5$ Using binomial theorem.

Ans ~~100~~ $(99)^5$

$$\text{Also } (100-1)^5 = {}^5C_0 + {}^5C_1 + {}^5C_2$$

$$+ {}^5C_3 + {}^5C_4$$

$$(100-1)^5 = {}^5C_0 (100)^{5-0} (-1)^0 + {}^5C_1 (100)^{5-1} (-1)^1 + {}^5C_2 (100)^{5-2} (-1)^2$$

$$+ {}^5C_3 (100)^{5-3} (-1)^3 + {}^5C_4 (100)^{5-4} (-1)^4 + {}^5C_5 (100)^{5-5} (-1)^5$$

$$= {}^5C_0 (100)^5 \cdot 1 + {}^5C_1 (100)^4 (-1) + {}^5C_2 (100)^3 \cdot 1 + {}^5C_3 (100)^2 (-1)$$

$$+ {}^5C_4 (100)^1 (1) + {}^5C_5 (100)^0 (-1)$$

$$= 1 (100)^5 - 5 (100)^4 + \frac{5!}{2!(5-2)!} (100)^3 - \frac{5!}{3!(5-3)!} (100)^2$$

$$+ \frac{5!}{4!(5-4)!} (100) - 1 (1).$$

$$= (100)^5 - 5 (100)^4 + 10 (100)^3 - 10 (100)^2 + 5 (100) - 1$$

$$= (100)^5 + 10 (100)^3 + 5 (100) - 5 (100)^4 - 10 (100)^2 - 1$$



$$100000000 + 10000000 + 800$$

$$(10)^{10} + (10)^9 + 5(10)^8 - 5(10)^8 - 10^{59} - 1$$

$$= 10010000500 - 500100001$$

$$= \underline{\underline{9509900499}}$$

34

IMPORTANT-NOTES

$$\begin{array}{r} 10010000500 \\ 500100001 \\ \hline \end{array}$$

$$\underline{\underline{9509900499}}$$

Finding Terms

$$T_{r+1} = {}^nC_r a^{n-r} b^r \rightarrow \text{General Term}$$

$$T_3 = T_{2+1}$$

$$T_4 = T_{3+1}$$

making it in T_{r+1} form

Q. Find G.T for $(x^2 - y)^6$

Here

$$r = r, a = x^2, b = -y$$

$$n = 6$$

Now

Using

$$T_{r+1} = {}^nC_r a^{n-r} b^r$$

Putting value

$$T_{r+1} = {}^6C_r (x^2)^{6-r} (-y)^r$$

$$= \boxed{{}^6C_r x^{12-2r} (-y)^r}$$

$$(x^2 - yx)^{12}$$

Here $r=0$, $a=x^2$, $b=yx$, $n=12$.

Using

$$T_{r+1} = {}^nC_r a^{n-r} b^r$$

putting values.

$$\therefore T_{r+1} = {}^{12}C_r (x^2)^{12-r} (yx)^r$$

$$\Rightarrow {}^{12}C_r x^{24-2r} (yx)^r$$

$$\Rightarrow {}^{12}C_r x^{24-2r} (-y)^r (x)^r$$

$$\Rightarrow {}^{12}C_r x^{24-2r+r} (-y)^r$$

$$\Rightarrow {}^{12}C_r x^{24-r} (-y)^r$$

Q. $\left(\frac{x}{3} - 3y\right)^7$ find 5th term

Sol. ~~Here~~ Here

$$r=4, n=7, a=\frac{x}{3}, b=-3y$$

using

$$T_{r+1} = {}^nC_r a^{n-r} b^r$$

putting value

$$T_{4+1} = {}^7C_4 a^{7-4} b^4$$

$$\Rightarrow \frac{7!}{4!(7-4)!} \left(\frac{x}{3}\right)^3 (-3y)^4$$

$$\Rightarrow \frac{7 \times 6 \times 5 \times 4 \times 3!}{4! \times 3!} \frac{x^3}{27} 81y^4$$



$$\Rightarrow 35 \cdot \frac{x^3}{27} \cdot 81y^4$$

$$\begin{array}{r} 35 \times 81 \\ 35 \times 80 \\ \hline 2835 \end{array}$$

$$\begin{array}{r} 105 \times 81 \\ 105 \times 80 \\ \hline 8415 \end{array} \quad x^3 y^4 \quad \Rightarrow 105 x^3 y^4$$

Summations:

$$\sum_{n=1}^{n=5} n = 1+2+3+4+5$$

$$\sum_{n=1}^5 (n^3+1) = (1^3+1) + (2^3+1) + (3^3+1) + (4^3+1) + (5^3+1)$$

$$= 2 + 9 + 28 + 65 + 126 \dots$$

$$1+2+3+\dots+n \text{ terms} = \frac{n(n+1)}{2}$$

$$1^2+2^2+3^2+\dots+n^2 \text{ terms} = \frac{n(n+1)(2n+1)}{6}$$

$$1^3+2^3+3^3+\dots+n^3 \text{ terms} = \left(\frac{n(n+1)}{2} \right)^2$$

Sigma Notation and its properties

$$1 \Rightarrow y = \sum_{i=1}^k = k$$

$$2 \Rightarrow y = \sum_{i=1}^k 2 = 2 \sum_{i=1}^k = 2k$$

NOTE $\Rightarrow$ Summation notation can be separated in case of + and - signs.

$$y = \sum_{r=1}^n (r^2 + r + 3)$$

$$\Rightarrow \sum_{r=1}^n r^2 + \sum_{r=1}^n (r) + \sum_{r=1}^n 3$$

$$\Rightarrow \dots + \dots + 3 \sum_{r=1}^n$$

$$S = t_1 + t_2 + t_3 + t_4 + \dots + t_n$$

$$S = \sum_{k=1}^n t_k$$

- To find the sum of the 1st n Natural Numbers.

$$S = 1 + 2 + 3 + \dots + n$$

$$S = \sum_{k=1}^n k = \frac{n(n+1)}{2}$$

Q. If the sum of 'n' terms of a sequence be $(2n^2 + 4)$, find its n^{th} term. ~~The sequence is~~

$$S_n = (2n^2 + 4)$$

$$S_{n-1} = (2(n-1)^2 + 4)$$

$$t_n = S_n - S_{n-1}$$

$$= (2n^2 + 4) - (2(n-1)^2 + 4)$$

$$= 2n^2 + 4 - 2(n-1)^2 - 4$$

$$= 2n^2 - 2(n-1)^2$$

$$= 2(n^2 - (n-1)^2)$$

$$= 2(n + n-1)(n - n+1)$$

$$t_n = 2(2n-1)$$



we can get terms only from 1.

to get the 1st term.
we will get S_1 .

$$S_1 = 2(1)^2 + 4$$

$$= 2 + 4 = 6$$

$$\therefore t_1 = 6$$

$$t_2 = 2(2(2)-1) = 2(4-1) = 6$$

$$t_3 = 2(2(3)-1) = 2(6-1) = 10$$

$$t_4 = 2(2(4)-1) = 2(8-1) = 14$$

$$t_5 = 2(2(5)-1) = 2(10-1) = 18$$

$\therefore$ sequence $= 6 + 6 + 10 + 14 + 18 \dots$

Find the sum of n terms of the series
whose n^{th} term is $n(n-1)(n+1)$.

$$S = t_1 + t_2 + t_3 + \dots + t_n$$

$$t_n = n(n-1)(n+1)$$

$$t_k = k(k-1)(k+1)$$

$$= k(k^2 - 1)$$

$$\underline{t_k = (k^3 - k)}$$

$$\left[S_n = \sum_{k=1}^n t_k \right]$$

$$\textcircled{Q} \sum_{k=1}^n (k^3 - k)$$

$$= \sum_{k=1}^n k^3 - \sum_{k=1}^n k$$

$$= (1^3 + 2^3 + 3^3 + \dots + n^3) - (1 + 2 + 3 + \dots + n)$$

$$= \left(\frac{n(n+1)}{2} \right)^2 - \frac{n(n+1)}{2}$$

$$= \frac{n(n+1)}{2} \left[\frac{n(n+1)}{2} - 1 \right]$$

$$= \frac{n(n+1)}{2} \left[\frac{n(n+1) - 2}{2} \right]$$

$$= \frac{n(n+1)}{2} \left[\frac{n^2 + n - 2}{2} \right]$$

$$= \frac{n(n+1)}{2} \left(\cancel{n(n+1)} \right) \left[\frac{n^2 + 2n - n - 2}{2} \right]$$

$$= \frac{n(n+1)}{2} \left[\frac{n(n+2) \cdot (n-1)}{2} \right]$$

$$= \frac{n(n+1)}{2} \left[\frac{(n+2)(n-1)}{2} \right]$$



$$\frac{n(n+1)}{4} (n+2)(n-1)$$

$$= \frac{n(n+2)(n^2-1)}{4} \quad \underline{\underline{\text{Ans}}}$$

Q. If the sum of 'n' terms of sequence be $(2n^2+4)$, find its n^{th} term.

Sum of n terms $(S_n) = t_1 + t_2 + t_3 + \dots + t_n$.

$$t_n = 2n^2 + 4$$

$$t_k = 2k^2 + 4$$

$$\sum_{k=1}^n (2k^2 + 4) = \sum_{k=1}^n 2k^2 + \sum_{k=1}^n 4$$

$$\rightarrow 2 \sum_{k=1}^n k^2 + 4n$$

$$= 2 (1^2 + 2^2 + 3^2 + 4^2 \dots + n^2) + 4n$$

$$= 2 \left(\frac{n(n+1)(2n+1)}{3} \right) + 4n$$

$$= \frac{n(n+1)(2n+1)}{3} + 4n$$

$$\frac{n(n+1)(2n+1)}{3} + 12n$$

$$= \frac{n(2n^2 + n + 2n + 1) + 12n}{3}$$

$$= \frac{n(2n^2 + 3n + 1) + 12n}{3}$$

$$= \frac{2n^3 + 3n^2 + n + 12n}{3}$$

$$= \frac{2n^3 + 3n^2 + 13n}{3}$$

Q Find the sum of 80 terms of the series whose n^{th} term is $n(n^2 - 1)$.

$$S_n = t_1 + t_2 + t_3 + \dots + t_{80}$$

$$t_n = n(n^2 - 1)$$

$$\sum_{k=1}^{80} (k^3 - k)$$

$$t_k = k(k^2 - 1)$$

$$\boxed{t_k = k^3 - k}$$

$$\sum_{k=1}^{80} k^3 - \sum_{k=1}^{80} k$$

$$S_{80} = \sum_{k=1}^{80} t_k$$

$$\Rightarrow (1^3 + 2^3 + 3^3 + 4^3 + \dots + 80^3) - (1 + 2 + 3 + \dots + 80)$$



$$S_n = \left(\frac{n(n+1)}{2} \right)^2 - \left(\frac{n(n+1)}{2} \right)$$

put $n = 40$

$$= \left(\frac{40 \times 81}{2} \right)^2 - \left(\frac{40 \times 81}{2} \right)$$

$$(40 \times 81)^2 - (40 \times 81)$$

$$= \underline{\underline{40 \times 81 (40 \times 81 - 1)}}$$

$$= 3240 (3240 - 1)$$

$$= \underline{\underline{3240(3239)}}$$

Category B

Find the sum of the following series.

$$1^3 + 3^3 + 5^3 + \dots \text{ to } n \text{ terms.}$$

Let take series as

$$1 + 3 + 5 + \dots + n.$$

$\therefore$ It is A.P. as C.D is 2

$$\therefore a_n = a + (n-1)d = 1 + (n-1)2 = 1 + 2n - 2 = (2n - 1)$$

the n^{th} term of the series

$$1^3 + 3^3 + 5^3 + \dots + \text{the } n^{\text{th}} \text{ term is } (2n-1)^3$$

$$\therefore t_n = (2n-1)^3$$

$$8n^3 - 1 - 12n^2 + 6n$$

$$t_k = \sum_{k=1}^n t_k$$

$$= \sum_{k=1}^n (8n^3 - 1 - 12n^2 + 6n)$$

$$= \sum_{k=1}^n 8k^3 - \sum_{k=1}^n 1 - 12 \sum_{k=1}^n k^2 + 6 \sum_{k=1}^n k$$

$$= 8 \sum_{k=1}^n k^3 - n - 12 \sum_{k=1}^n k^2 + 6 \sum_{k=1}^n k$$

$$= 8 \left(\frac{n(n+1)}{2} \right)^2 - n - 12 \left(\frac{n(n+1)(2n+1)}{6} \right)$$

$$+ 6 \frac{n(n+1)}{2}$$

$$= 2(n(n+1))^2 - n - 2(n(n+1)(2n+1)) + 3(n(n+1))$$



$$2(n(n+1))^2 - n - 2n(n+1)(2n+1) + 3n^2 + 3n$$

$$= 2(n(n+1))^2 - n - 2n(2n^2 + n + 2n + 1) + 3n^2 + 3n$$

$$= 2(n(n+1))^2 - n - 2n(2n^2 + 3n + 1) + 3n^2 + 3n$$

$$= 2(n^2(n^2 + 2n + 1)) - n - 2n(2n^2 + 3n + 1) + 3n^2 + 3n$$

$$= 2(n^4 + 2n^3 + n^2) - n - 4n^3 - 6n - 1 + 3n^2 + 3n$$

$$= 2n^4 + 4n^3 + 2n^2 - n - 4n^3 - 6n - 1 + 3n^2 + 3n$$

$$= 2n^4 + 4n^3 + 2n^2 - 4n - 1$$

$$= 2n^3(n+1)$$

$$2n^4 + 4n^3 + 2n^2 - n - 4n^3 - 6n - 1 + 3n^2 + 3n$$

$$= 2n^4 + 5n^2 - 4n - 1$$

$$= n^2(2n^2 + 5) - 4n - 1$$

Linear Recurrence Relation with constant coefficients. **IMPORTANT NOTES**

→ Recurrence Relation.

- It is an equation that recursively defines a sequence where the next term is a function of the previous terms.

eg $S = \{2, 2^2, 2^3, \dots, 2^n\}$.

$S = \{S_n\} = \{2^n\} \rightarrow \text{sequence.}$

Recurrence relation $\rightarrow S_n = 2S_{n-1}, n \geq 2, S_1 = 2 \rightarrow \text{initial condition.}$

" $S_n = 2S_{n-1}, n \geq 2$ is recurrence relation with initial condition $S_1 = 2$.

Ex₂

$S = \{3, 7, 11, 15, 19, \dots\}$

$S_n = S_{n-1} + 4, n \geq 2, S_1 = 3.$

" $S_n = S_{n-1} + 4, n \geq 2$ is recurrence relation with initial condition $S_1 = 3$.

3. $S = \{1, 1, 2, 3, 5, 8, 13, \dots\}$

$S_n = S_{n-1} + S_{n-2}, n \geq 3$ with $S_1 = S_2 = 1$

Fibonacci Sequence.



$$C_0 a_n + C_1 a_{n-1} + C_2 a_{n-2} + \dots + C_k a_{n-k} = f(n)$$

where $C_0, C_1, C_2, \dots, C_k$ are constants.

$$\text{ex } 2a_n + 3a_{n-1} = 3 \checkmark$$

$$a_n - 7a_{n-1} + 12a_{n-2} = \underline{n-4^n} \checkmark$$

⑧

Order: It is the difference between the highest and lowest subscript of a .

$$\text{ex } 2a_n + 3a_{n-1} = 3$$

$$\begin{aligned} \text{order} &= n - (n-1) \\ &= n - n + 1 = 1. \end{aligned}$$

$$\text{ex } a_n + a_{n-1} + a_{n-2} + a_{n-3} = 0$$

$$\begin{aligned} \text{Order} &= n - (n-3) \\ &= n - n + 3 \\ &= 3. \end{aligned}$$

Degree: It is defined as the highest power of a_n .

Degree of linear recurrence relation is 1.

The method of Characteristic Roots.

Linear Rec. Relation with const. coeff.

$$[C_0 a_n + C_1 a_{n-1} + \dots + C_k a_{n-k} = b(n)]$$

Homogeneous
Recurrence Relation

$$b(n) = 0$$

Non-homogeneous
Recurrence Relation

$$b(n) \neq 0$$

Homogeneous Recurrence relation.

Steps to solve.

1. Put $a_n = r^n$, $a_{n-1} = r^{n-1}$ and so on.
2. Find an equation in terms of r . This is called as characteristic equation or Auxiliary equation.
3. solve characteristic equation and the characteristic roots.
4. If characteristic roots are.

a) $r = r_1, r_2, r_3$ (distinct)

general solution is $a_n = b_1 r_1^n + b_2 r_2^n + b_3 r_3^n$

$b_1, b_2, b_3 \rightarrow \text{constants}$

b) $r = r_1, r_1$

$$\rightarrow a_n = (b_1 + n b_2) r_1^n$$

c) $r = r_1, r_1, r_2, r_3$



$$\rightarrow a_n = (b_1 + nb_2)r_1^n + b_3r_2^n + b_4r_3^n$$

$$d) \quad r = r_1, r_1, r_1, r_2$$

$$a_n = (b_1 + nb_2 + n^2b_3)r_1^n + b_4r_2^n$$

Q. Solve $a_n = a_{n-1} + 2a_{n-2}$, $n \geq 2$ with the initial condition $a_0 = 0$, $a_1 = 1$

$$a_n - a_{n-1} - 2a_{n-2} = 0$$

put $a_n = r^n$

$$r^n - r^{n-1} - 2r^{n-2} = 0$$

$$r^n \left[1 - \frac{1}{r} - \frac{2}{r^2} \right] = 0 \quad \text{or} \quad r^n [r^2 - r - 2] = 0$$

$$\Rightarrow r^2 - r - 2 = 0$$

$$\Rightarrow \underline{\underline{r = 2, -1}}$$

$$\Rightarrow a_n = b_1(2)^n + b_2(-1)^n$$

$$a_0 = 0 \Rightarrow 0 = b_1 + b_2 \quad \text{--- (I)}$$

$$a_1 = 1 \Rightarrow 1 = b_1(2) + b_2(-1)$$

$$1 = 2b_1 - b_2 \quad \text{--- (II)}$$

from ① & ② $b_1 = \frac{1}{3}$, $b_2 = \frac{1}{3}$.

$$\therefore a_n = \left(\frac{1}{3}\right)2^n + \left(-\frac{1}{3}\right)(-1)^n.$$

Q₂ Solve $a_n = 4(a_{n-1} - a_{n-2})$ with initial condition $a_0 = a_1 = 1$.

$$a_n - 4a_{n-1} + 4a_{n-2} = 0.$$

put $a_n = r^n$.

$$r^n - 4r^{n-1} + 4r^{n-2} = 0.$$

$$\Rightarrow r^n \left[1 - \frac{4}{r} + \frac{4}{r^2} \right] = 0.$$

$$\Rightarrow r^n [r^2 - 4r + 4] = 0.$$

$$\Rightarrow r^2 - 4r + 4 = 0$$

$$\Rightarrow r^2 - 2r - 2r + 4 = 0$$

$$\Rightarrow r(r-2) - 2(r-2) = 0$$

$$= (r-2)(r-2)$$

$$\Rightarrow r = 2, 2.$$

$$a_n = (b_1 + nb_2)r_1^n$$

$$\Rightarrow a_n = (b_1 + nb_2)2^n.$$



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$$a_0 = 1 \Rightarrow (b_1 + 0(b_2))2^0 = (b_1)1 = 0b_1$$

$$2b_1 \rightarrow b_1 = \frac{1}{2}$$

$$a_1 = 1 = (b_1 + 1(b_2))2^1 = (b_1 + b_2)2$$

$$\Rightarrow 1 = 2b_1 + 2b_2$$

$$\Rightarrow 1 = 2\left(\frac{1}{2}\right) + 2b_2$$

$$1 = 1 + 2b_2$$

$$\Rightarrow 2b_2 = 1 - 1$$

$$\Rightarrow b_2 = 0$$

$$\boxed{b_1 = 1}$$

$$a_1 = 1 = (b_1 + 1b_2)2^1 = 2b_1 + 2b_2$$

$$1 = 2 + 2b_2$$

$$1 - 2 = 2b_2$$

$$\boxed{b_2 = -\frac{1}{2}}$$

$$a_n = \left(1 + n\left(-\frac{1}{2}\right)\right)2^n$$

$$\Rightarrow \left(1 - \frac{n}{2}\right)2^n$$

PROPOSITION $\Rightarrow$ It is a declarative sentence (a sentence that is declaring a fact or stating an argument) which can be either TRUE or FALSE but cannot be both.

e.g.

- 1) Delhi is the capital of India.
- 2) Water Froze this morning.
- 3) $1+1=2$.

Sentences which are not Propositions.

- 1) What time is it?
- 2) $x+1=2$ (can be both True or False)

Propositional Logic.

Area of logic that studies ways of joining and/or modifying propositions to form more complicated propositions and it also studies the logical relationships and properties derived from these combined / altered propositions.

e.g.

Statement 1 :- "Adam is good in playing football".

Statement 2 :- "Adam is good in playing football and this time he is representing his college at national level".

Proposition 2

Proposition 1

joining
two propositions
with logical
connective

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NOTE → Propositional Logic is sometime called as

IMPORTANT NOTES

"Sentential logic" or "statement logic".

Propositional Variables:

Variables that are used to represent propositions are called propositional variables.

if p = Adam is good in playing football.

q = This time he is representing his college at National level.

$$p \wedge q \longrightarrow p \text{ and } q$$

Logical operators

There are 6 logical operators that we will focus on:-

- 1) Negation
- 2) Conjunction
- 3) Disjunction
- 4) Exclusive OR
5. Implication
6. Biconditional.

1) Negation:

Its symbol is $\neg$.

Let p be a proposition. $\neg p$ is called negation of p which simply states that "It is not the case that p ".

if p is true then $\neg p$ is false. if p is false then $\neg p$ is true.

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Example: let p be a proposition -

"Adam and Eve lived together for many years".

"It is not the case that Adam and Eve lived together for many years."

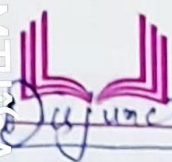
p	$\neg p$
T	F
F	T

Let p and q be two propositions. Conjunction of p and q is denoted by $p \wedge q$.. when both p and q are true then only the compound proposition $p \wedge q$ is true.

Truth table

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Important note
↳ Sometimes we use "but" instead of "and."



Disjunction. (Inclusive - OR)

Let p and q be propositions. Disjunction of p and q is denoted by $p \vee q$. When both p and q are false then only the compound proposition $p \vee q$ is false.

ex $\underbrace{16 - 4 = 10}_F \text{ or } \underbrace{4 \text{ is an even number}}_T$

Truth table

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Exclusive - OR.

Let p and q be two propositions. The exclusive OR of p and q (denoted by $p \oplus q$) is a proposition that simply means exactly one of p and q will be true but both cannot be true.

Truth Table:

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

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Implication.

Let p and q be propositions. The proposition "if p then q " denoted by $p \rightarrow q$ is called implication or conditional statement.

Truth table

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

p is called hypothesis (or premise) and q is called conclusion (or consequence).

Ex 1 "If you try hard for your exam, then you will succeed."

Here p = you tried hard for your exam.
 q = you succeed.

Case 1 : "You tried hard for your exam" and "you succeed"
 $p = \text{true}$ $q = \text{True.}$

Compound proposition $p \rightarrow q$ is True.

Case 2 : "You tried hard for your exam" but "you failed";
 $p = \text{true}$ $q = \text{false.}$

Compound proposition $p \rightarrow q$ is False.

Case 3 : "You haven't tried hard for your exam" and "You succeed".
 $p = \text{false}$ $q = \text{true.}$

Compound proposition $p \rightarrow q$ is True

Why? because you make the compound proposition false only when you satisfy the first condition itself i.e., p . If that itself not satisfied

we cannot make compound proposition false,
~~if~~ if false means True.

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"You haven't tried hard for your exam" and
 $p = \text{False}$

"You failed"

$q = \text{False}$

Compound proposition $p \rightarrow q$ is true.

Note $\rightarrow$ If p is false then it doesn't matter what will be
 the truth value of $p \rightarrow q$ is always TRUE.

Representation of Conditional Statements

Different ways to represent conditional statements:

"if p then q "

" p only if q "

" p implies q "

" q is necessary for p "

" q when p "

" p is sufficient for q "

" q whenever p "

" q unless $\neg p$ "

" q follows from p "

" p and if q "

Ex "I will stay at home only if I'm sick".

Let $p =$ "I will stay at home" and let $q =$ "I'm sick".

According to the above statement, becoming sick
 is the necessary condition that will make you
 stay at home.

This means "if you are not sick then, you cannot stay at home at any cost".

IMPORTANT-NOTES

In order to falsify the above statement, q must be false and p must be true i.e., you are not sick and you still stay at home.

Proof idea $\Rightarrow$ Truth value of p and q must be same in order to falsify the statement.

if p then q $\Rightarrow$ "If I'll stay at home then I'm sick".

Then only way to falsify the above statement is by making p True and q false. Therefore, only if q is equivalent to if p then q .

" q unless $\neg p$ "

$\hookrightarrow$ means 'except if'

"if p then q "



" q is true when p is true"



" q is true except when p is false"



" q is true unless p is false" = " q unless $\neg p$ ".



Implication or Conditional statement: $P \rightarrow Q$

Converse = $Q \rightarrow P$

Contrapositive = $\neg Q \rightarrow \neg P$

Inverse = $\neg P \rightarrow \neg Q$

Example

"If it rains today then, I will stay at home"

Converse: If I will stay at home then it rains today

Contrapositive: If I will not stay at home, then it does not rain today.

Inverse: ~~I will not stay at home~~
If it does not rain today then, I will not stay at home.

Facts

- Implication and Contrapositive both are equivalent.
- Converse and Inverse both are equivalent.
- Neither Converse nor inverse is equivalent to Implication.

Truth table

IMPORTANT-NOTES

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$q \rightarrow p$	$\neg q \rightarrow \neg p$	$\neg p \rightarrow \neg q$
T	T	F	F	T	T	T	T
T	F	F	T	F	T	F	T
F	T	T	F	T	F	T	F
F	F	T	T	T	T	T	T

Biconditional

Let p and q be 2 propositions. The biconditional statement of the form:

$p \leftrightarrow q$ is the proposition "p if and only if q"

$p \leftrightarrow q$ is true whenever the truth value of p and q are same.

Truth table

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

How "p if and only if q" makes sense?

"p if and only if q" composed of two statements -

"p if q" and "p only if q".

"p only if q" = "if p then q"

and

"p if q" = "if q then p"

$$(p \rightarrow q) \wedge (q \rightarrow p) = p \leftrightarrow q$$

Representation

1. p is necessary and sufficient for q and vice versa
2. if p then q , and conversely.
3. p iff q .

Example: let p be a proposition "You get promoted" and let q be a proposition "You have connections" then $p \leftrightarrow q$ is the statement: "You get promoted if and only if you have connections."

Tautology

It is a compound proposition which is always true.

ex

p	$\neg p$	$p \vee \neg p$
T	F	T
F	T	T

Contradiction

Compound proposition which is always False.

ex

p	$\neg p$	$p \wedge \neg p$
T	F	F
F	T	F

Contingency:

Compound proposition which is sometimes True and sometimes False.

eg

P	q	$P \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Satisfiability:

A compound proposition is satisfiable if there is at least one TRUE result in its truth table.

Tautology is always satisfiable. Contingency is satisfiable.

Unsatisfiability: not even a single TRUE result in its truth table.

A Contradiction is always unsatisfiable.

Valid: A compound proposition is valid when it is a tautology.

Invalid: A compound proposition is invalid when it is either a contradiction or contingency.

→ Tautology is always satisfiable. **IMPORTANT-NOTES**
always tautology.

→ Invalid not only mean a compound proposition is always False. If a compound proposition is sometimes TRUE and sometimes FALSE, then also it is said to be invalid.

LOGICAL Equivalences. (Laws)

The compound proposition p and q are said to be logically equivalent if $p \leftrightarrow q$ is a Tautology. Logical equivalence is denoted by $\equiv$ or $\Leftrightarrow$.

Ex $\Rightarrow p \wedge T \equiv p$

p	T	$p \wedge T$
T	T	T
F	T	F

1) Identity Laws:

a) $p \wedge T \equiv p$

b) $p \vee F \equiv p$

2) Domination Laws:

a) $p \vee T \equiv T$

b) $p \wedge F \equiv F$

3) Idempotent Laws:

a) $p \vee p \equiv p$

b) $p \wedge p \equiv p$

4) Double Negation Law:

$$\neg(\neg p) \equiv p$$

5) Commutative Laws:

$$a) p \vee q \equiv q \vee p$$

$$b) p \wedge q \equiv q \wedge p$$

6) Associative Laws:

$$a) (p \vee q) \vee r \equiv p \vee (q \vee r)$$

$$b) (p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$$

7) Distributive Laws:

$$\rightarrow p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

$$\rightarrow p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

8) De Morgan's Law:

$$a) \neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$b) \neg(p \vee q) \equiv \neg p \wedge \neg q$$

9) Absorption Laws:

$$a) p \vee (p \wedge q) \equiv p$$

$$b) p \wedge (p \vee q) \equiv p$$

Proof

$$p \vee (p \wedge q) = (p \vee p) \wedge (p \vee q) \quad \text{f using Distributive Law}$$

$$= p \wedge (p \vee q) \quad \{ \because p \vee p = p \}$$

$$= (p \wedge p) \vee (p \wedge q)$$

$$= p \vee (p \wedge q) \quad \{ \because p \wedge p = p \}$$



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$$= p \wedge (1 \vee q)$$

~~of Taking p as common~~ **IMPORTANT-N**

$$= p \wedge 1$$

$$\{ \because 1 \vee q = 1 \}$$

$$= p$$

$$\{ \because p \wedge 1 = p \}$$

Negation laws:

$$a) p \vee \neg p \equiv T$$

$$b) p \wedge \neg p \equiv F$$

LOGICAL EQUIVALENCE INVOLVING CONDITIONAL STATEMENTS

$$\underline{1} \bullet p \rightarrow q \equiv \neg p \vee q$$